

Customer Personality Analysis

Group 3

Problem Statement :

- Analyze the effectiveness of discounts on customer purchasing behavior.



Introduction (What Data Did we Possess?)

- Customer Demographics (ID, Year of Birth, Education, Marital Status, and Income)
- Household Compositions (KidHome, TeenHome)
- Purchasing Behavior (Amount Spent on Items & Discounted Deals Purchased)
- Promotional Campaigns (1 to 5 and Response Campaign)
- Shopping Channels

Methodology and Work Flow



Exploratory Data Analysis & Visualization

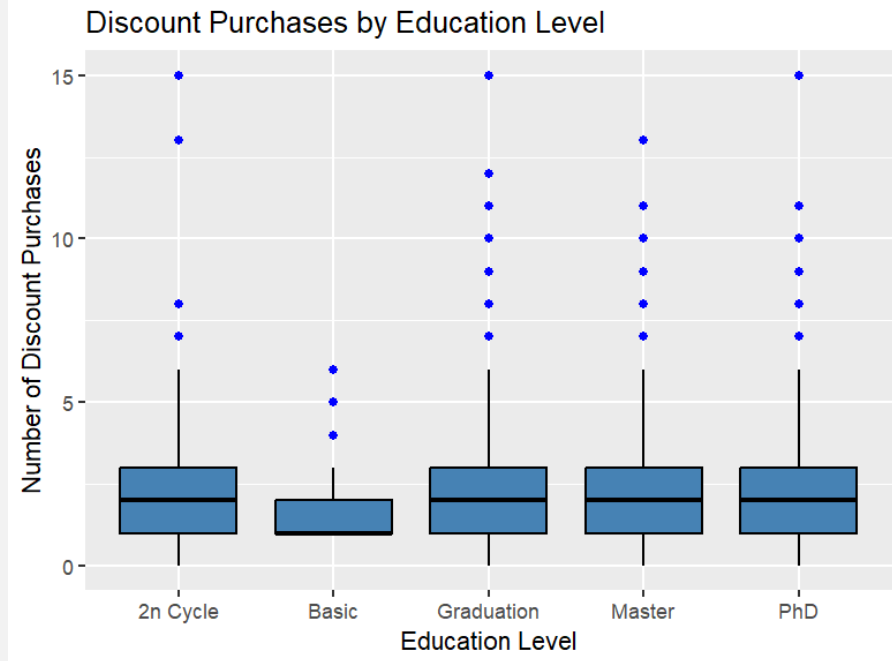


Customer Segmentation

Customers have been segmented on various factors to assess their purchasing behavior when it comes to discounts. Mostly they are dissected:

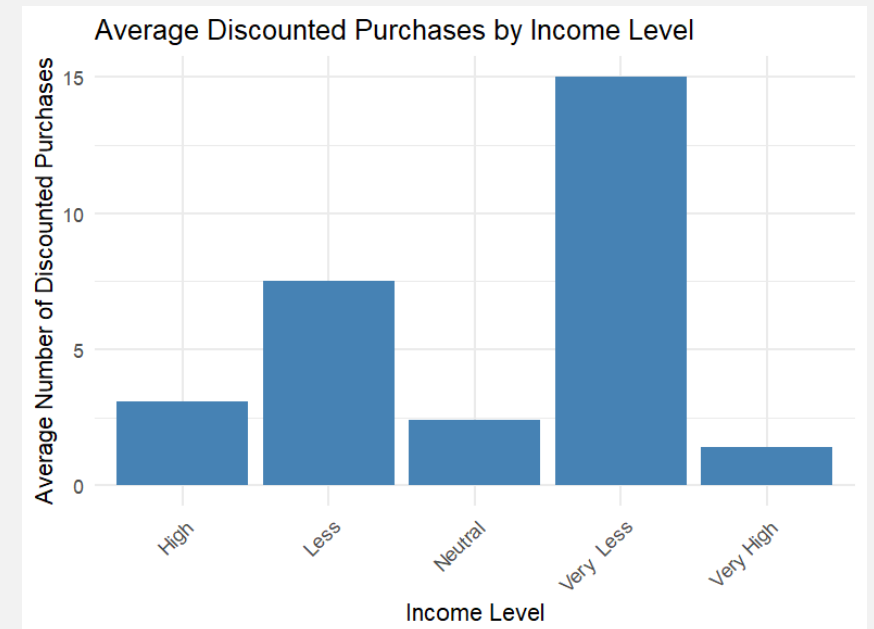
- Level of Income
- Level of Education
- Amount of Spending
- Discount Used or Not

Discount Purchases Based on Income Level & Education



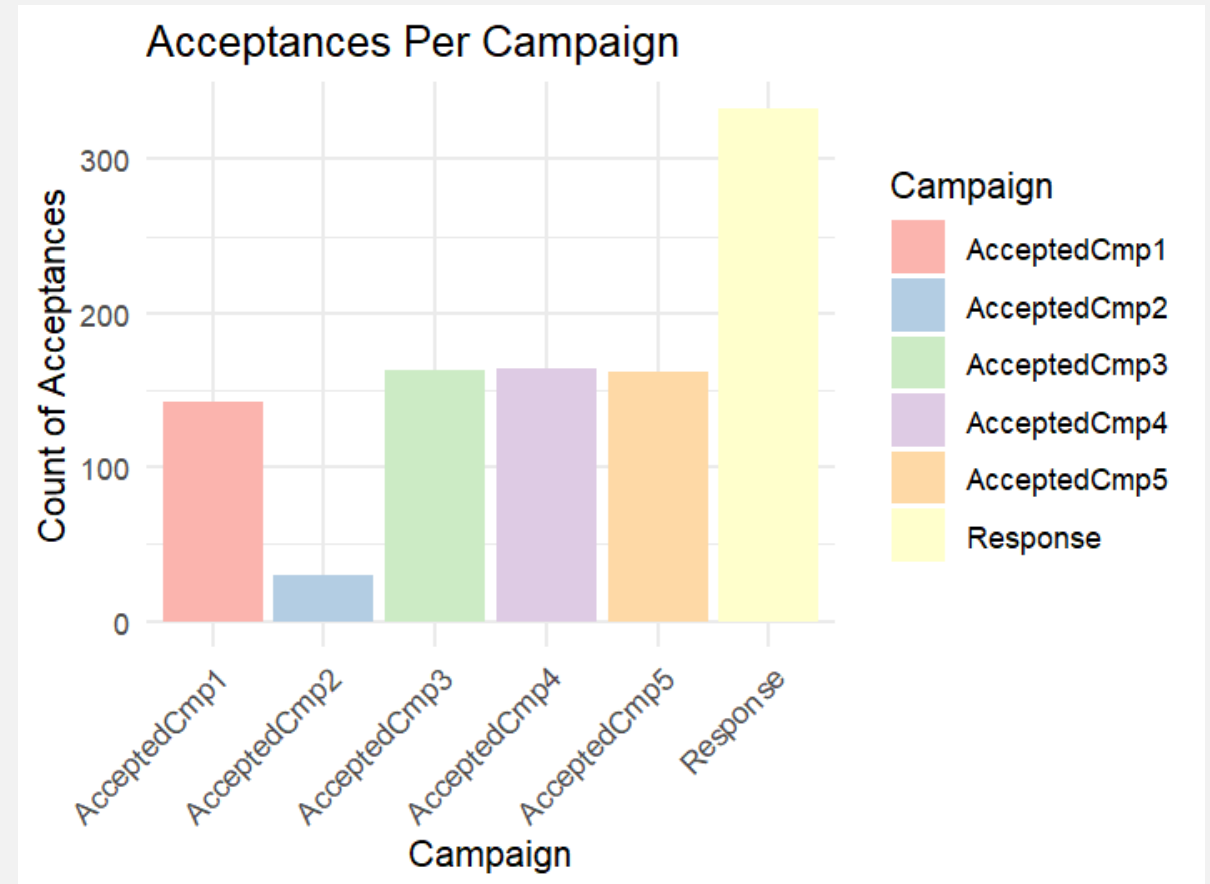
- Individuals with 'Master' and 'PhD' levels of education have a wider range of discount purchases, with several outliers indicating high numbers of discount purchases.
- '2n Cycle' and 'Basic' education levels show a more consistent, but lower number of discount purchases.

- The 'Very Less' income group shows the highest average number of discounted purchases, while the 'Very High' income group has the lowest average.
- The 'High' and 'Very High' income groups show significantly fewer average discounted purchases than the 'Less' group.



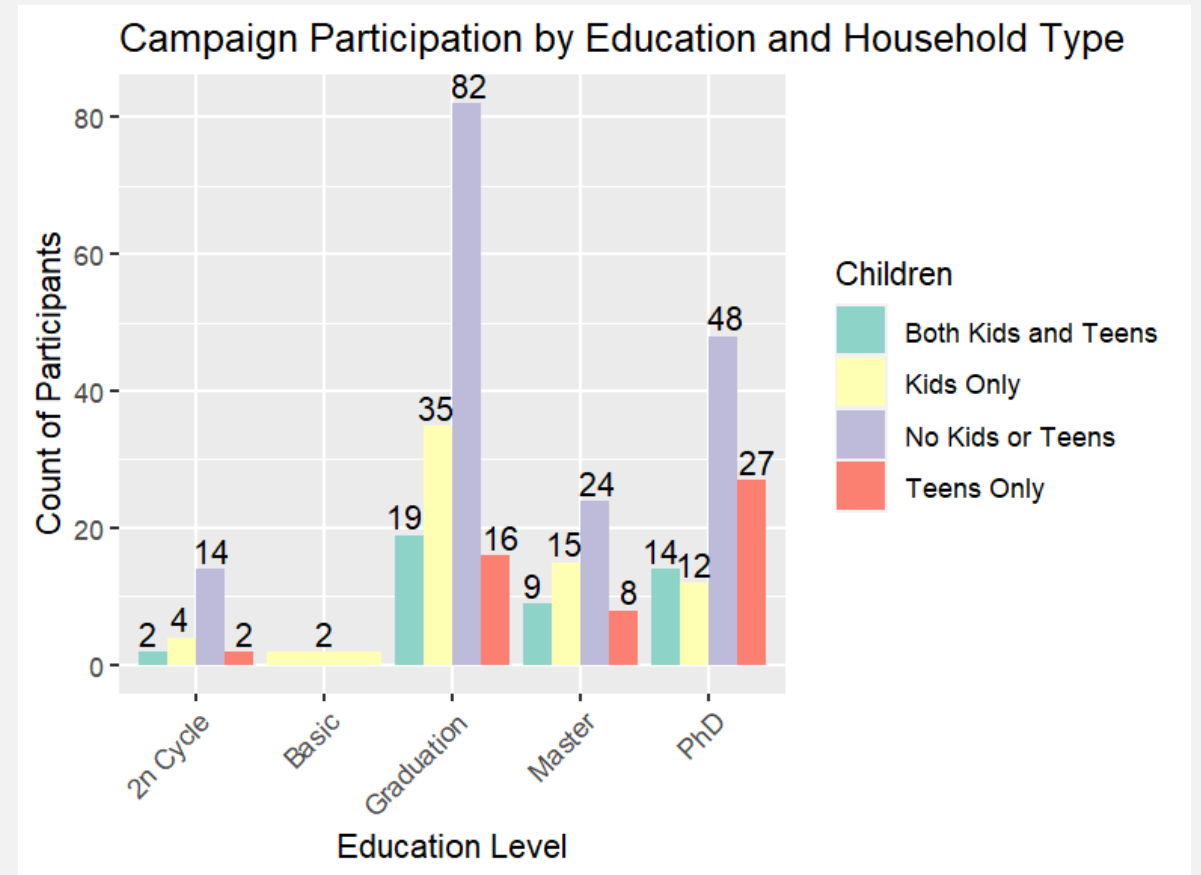
Most Successful Promotional Campaign

- The Response Campaign stood out as the winner as it resonated with the audience.
- The second campaign significantly underperformed in comparison to its counterparts, marking it as the least engaging initiative.
- Notable variations across Campaigns 1, 3, 4 & 5 suggesting varying degrees of alignment with customer interests and preferences.



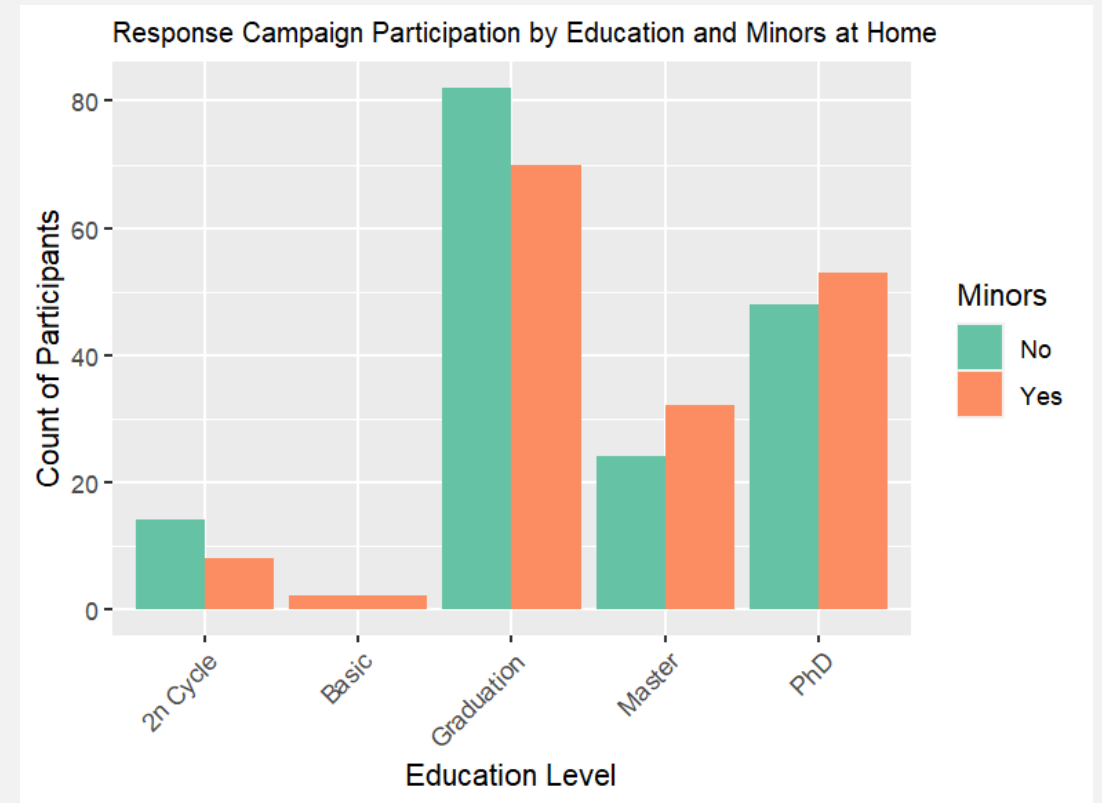
Campaign Participation By Education and Household Type

- Graduate Customers who don't have kids saw the most number of participants for Campaigns.
- Customers with No Kids or Teens in general participated more in campaigns
- Very few '2n Cycle' and 'Basic' level participants joined the campaign, regardless of household type.



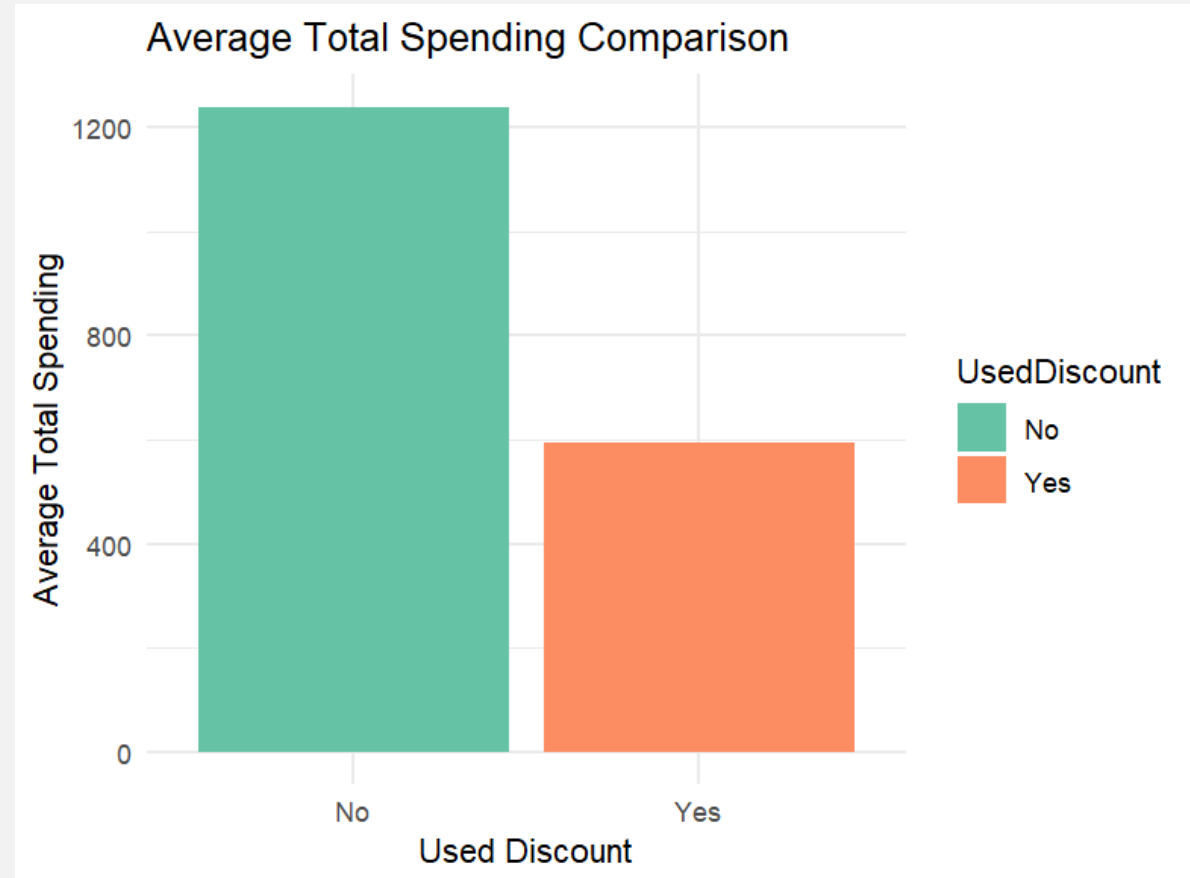
Educated Customers Purchase Using Campaigns?

- Most participants who joined the 'Response' campaign have a 'Graduation' level education.
- Having kids at home didn't consistently affect whether people participated in the campaign.
- Fewer people with 'Basic' or '2n Cycle' education levels took part in the campaign.



Discounted Purchases Vs Non Discounted

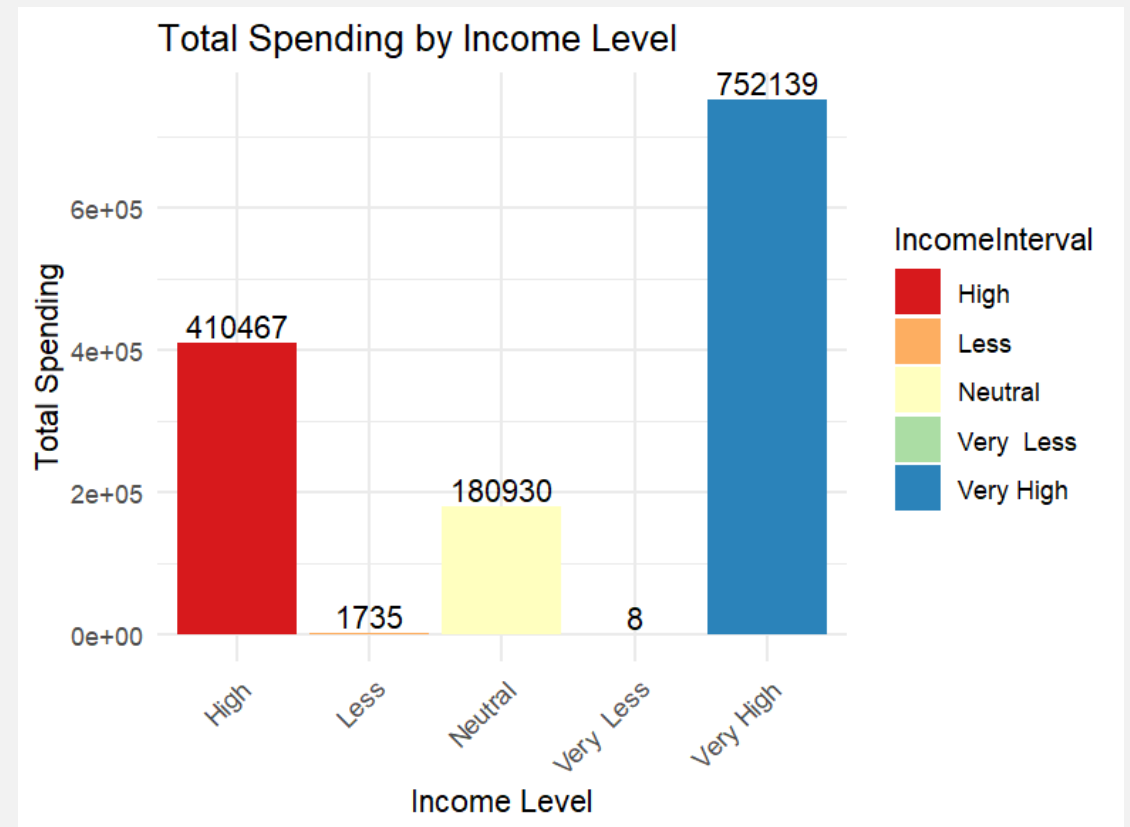
Used Discount	No	Yes
Mean	\$12.38M	\$5.94M
Median	\$13.56M	\$3.86M
Standard Deviation	\$7.97M	\$5.91M
Frequency of Purchase	5.57	5.79



Do the Rich Care About Discounts?

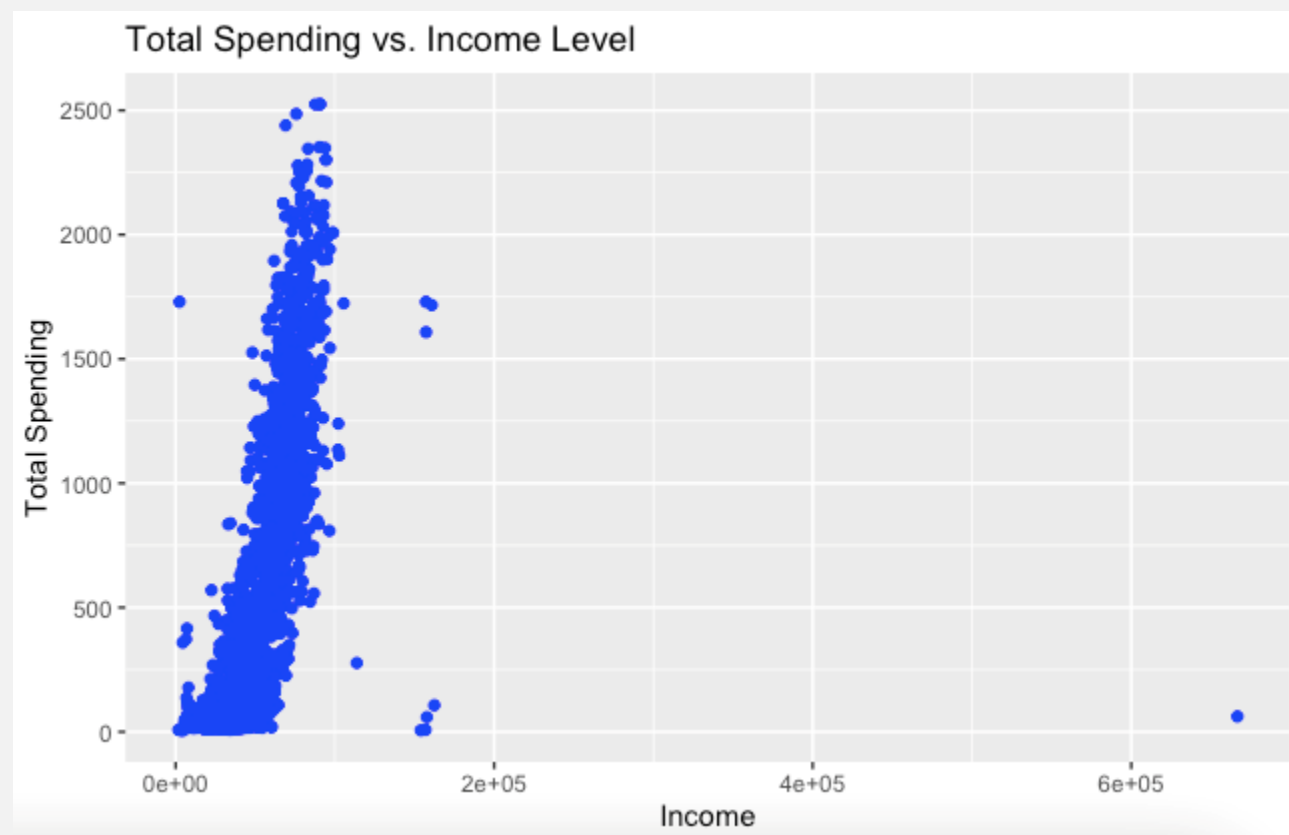
From the previous slide

- Customers with above-average income levels spend on items regardless of discounts
- The 'Less', 'Neutral', and 'Very Less' income groups contribute much less to the total spending mostly from discounts that too.



What kind of correlation can we see?

- Income Level and Total Spending seem to have a positive correlation
- As income increases the spending increases as well.
- This corroborates our previous point that customers with high-income levels spend more regardless of discount



Insights & Strategic Recommendations



Insights

EDA Employed on Questions	Key Insight Drawn
<u>Discount Purchases Based on Income Level & Education</u>	• The higher variability in the 'Master' and 'PhD' categories may suggest that customers with higher education levels are more likely to take advantage of discounts for larger purchases. • The 'High' and 'Very High' income groups' lower average might suggest they are less motivated by discounts or that the discounts are not on products they typically purchase.
<u>Most Successful Promotional Campaign</u>	• The success of the 'Response' campaign may warrant a deeper dive to uncover the key drivers behind its outstanding performance
<u>Campaign Participation By Education and Household Type</u>	• Graduate Level customers mainly resonate with the campaigns that are advertised and mainly with no minors at home

Insights

EDA Employed on Questions	Key Insight Drawn
<u>Educated Customers Purchase Using Campaigns?</u>	• The high participation of customers with a 'Graduation' education level may reflect campaign content that aligns well with the values or interests of this group.
<u>Do the rich care about Discounts?</u>	• The spending power of the 'Very High' income group suggests that they may not be as price-sensitive and could be targeted with premium offerings. • Lower income groups may benefit from more aggressive discounts or value-focused marketing to boost their spending.
<u>Correlation Between Income & Total Spending</u>	• The spread of spending at lower income levels might reflect budget constraints or a selective purchasing approach focused on essential or discounted items. • The trend of increased spending with higher income supports the common economic theory that as disposable income rises, so does consumer spending.

Strategic Recommendations

- Tailor discount strategies to appeal more to the spending habits of customers with higher education, perhaps with offers on premium products.
- Focus on creating more compelling discount offers for the 'Very Less' income group to encourage increased purchasing.
- For the 'Less', 'Neutral', and 'Very Less' income groups, consider loyalty programs, special discounts, and promotions to increase engagement and spending.
- Focus on tending to High-Income customers since they are not driven by discounts.
- Develop a family-focused promotional campaign to attract customers with minors at home.
- Remove or try to improve promotional campaign 2 as it underperformed. Try to gain customer feedback and

Thank you

